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← Text to Numeric →

1. Bag of Words (BoW)
2. Binary BoW

} n-grams

3. TF-IDF
4. Word2Vec

Document → SMS
→ Tweets
→ Article

Corpus – Entire data set

D1: The car is driven on the road
D2: The truck is driven on the highway
⋮

Vocabulary
– the unique words in the corpus

Feature Extraction

1. Bag of Words:

	The	car	truck	is	driven	on	road	highway
	1	1	0	1	1	1	1	0
D1:	2	1	0	1	1	1	1	0
D2:	2	0	1	1	1	1	0	1
	1	0	1	1	1	1	0	1
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Binary BoW's

← Vocab →

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
1	2	1	0	1	1	1	1	0
2	2	0	1	1	1	1	0	1
⋮								
n								

Dino

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iphone is efficient but not cheap
oppo is cheap but not efficient → Trigram

	iphone	oppo	is	efficient	cheap	but	not
D1	1	0	1	1	1	1	1
D2	0	1	1	1	1	1	1

n - gram -
1 → unigram
2 → Bigram
3 → Trigram

D1. The car is driven on the road
D2: The truck is driven on the highway → Bigram.

	X_1 The car	X_2 The truck	X_3 car is	X_4 truck is	X_5 is driven	X_6 driven on	X_7 on the	X_8 the road	X_9 the highway
D1:	1	0	1	0	1	1	1	1	0
D2	0	1	0	1	1	1	1	0	1

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TF-IDF

Term Frequency ↵ ↳ Inverse Document Frequency

— Words are given Values based on their importance in the document

$$\text{Term Frequency (TF)} = \frac{\#\text{ Term occurs}}{\#\text{ Total No. of words in the doc}}$$

$$\text{Document Frequency (DF)} = \frac{\#\text{ Docs in which word occurs}}{\#\text{ Total Documents}}$$

$$IDF = \frac{1}{DF}$$

3 / 4

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D1: The car is driven on the road (7)
D2: The truck is driven on the highway (7)

	Term Freq D1	Term Freq D2	Doc Freq	IDF	log IDF	TF-IDF D1	TF-IDF D2
The	2/7	2/7	2/2	1	0	0	0
car	1/7	0	1/2	2	0.30	0.043	0
truck	0	1/7	1/2	2	0.30	0	0.043
is	1/7	1/7	2/2	1	0	0	0
driven	1/7	1/7	2/2	1	0	0	0
on	1/7	1/7	2/2	1	0	0	0
road	1/7	0	1/2	2	0.30	0.043	0
highway	0	1/7	1/2	2	0.30	0	0.043

R1 → +ve
R2 → −ve
R3 →
⋮
Rm →

9:30 pm

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
1	0	0.043	0	0	0	0	0.043	0
2	0	0	0.043	0	0	0	0	0.043

Talking:

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1	C u soon	Ham
2	Win a trip for 2	spam

I/p → [BoW → Nv → SVC] → Ham/spam.

← Pipeline →

Raw Msg → [Tokenize → pos → NER → Text Nv → Clf Mod] →

Preprocessing

Reduce Vocab

1. Convert into lower case ✓
2. Remove stop words ✓
3. Stemming / Lemmatizing ✓

Parts of speech

I → Pro sub
Ive → Verb
ice cream → Object Noun

Fairly → Fair
Easily → Easi

4. Tokenizing ✓
5. pos Tagging ✓
6. NER — Named Entity Recognition ✓

Apple APPLE apple → 3
apple apple apple → 1.

NLTK →
spacy → [-]

run running runs ran — 4
— 1
run

Nouns → Not cut
runner → runner

Fairly → Fair
Boys → Boy

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Huge amt of data → Model →

Spacy ✓
NLTK
gensim

95% → Pre trained Models
↓
Fine-tune
↓
Medical / Legal

6 / 6